

Question 1

Use a direct proof to show that the sum of two even integers is even

Let a and b be two even integers. Then, by definition, there exist integers m and n such that $a = 2m$ and $b = 2n$.

Then, the sum of a and b is:

$$2m + 2n = 2(m + n)$$

Since integers are closed under addition, $m + n$ is an integer. Therefore, $2(m + n)$ is even.

Thus, the sum of two even integers is even. ■

Question 2

Show that the additive inverse, or negative, of an even number is an even number using a direct proof.

Let a be an even integer. Then, by definition, there exists an integer m such that $a = 2m$.

The additive inverse of a is $-a$.

Then, $-a = -2m = 2(-m)$

Since integers are closed under multiplication, $-1 \cdot m$ is an integer and $2(-m)$ is therefore even.

Therefore, the additive inverse of an even number is an even number.

■

Question 3

Use a direct proof to show that the product of two odd numbers is odd.

Let a and b be two odd integers. Then, by definition, there exist integers m and n such that $a = 2m + 1$ and $b = 2n + 1$.

Then, the product of a and b is:

$$(2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1$$

Since integers are closed under addition and multiplication, $2mn + m + n$ is an integer and $2(2mn + m + n)$ is therefore even. Thus, $2(2mn + m + n) + 1$ is odd.

Therefore, the product of two odd numbers is odd. ■

Question 4

Prove that if n is a perfect square, then $n + 2$ is not a perfect square.

Let n be a perfect square. Then, there exists an integer m such that $n = m^2$.

Assume for the sake of contradiction that the conclusion (" $n + 2$ is not a perfect square") is false. In that case, there exists an integer k such that $n + 2 = k^2$.

Substitute $n = m^2$ into the equation $n + 2 = k^2$:

$$m^2 + 2 = k^2$$

Rearranging the equation:

$$k^2 - m^2 = 2$$

This can be factored as:

$$(k + m)(k - m) = 2$$

Since k and m are integers, the only possible factor pairs of 2 are (1, 2) and (-1, -2).

Therefore, the only possible solutions for k and m are (1, 1) and (-1, -1).

However, if $k = 1$ and $m = 1$, then $n + 2 = 1^2 + 2 = 3$, which is not a perfect square.

Likewise, if $k = -1$ and $m = -1$, then $n + 2 = (-1)^2 + 2 = 3$, which is not a perfect square.

Therefore, if n is a perfect square, then $n + 2$ is not a perfect square.



Question 5

Prove or disprove that the product of a nonzero rational number and an irrational number is irrational.

Let n be a nonzero rational number. Then, there exist integers p and q such that $n = \frac{p}{q}$, where $q \neq 0$.

Let m be an irrational number.

Assume for the sake of contradiction that the product of n and m is rational. Then, there exist integers a and b such that $n \cdot m = \frac{a}{b}$, where $b \neq 0$.

In the equation $n \cdot m = \frac{a}{b}$, substitute $n = \frac{p}{q}$:

$$\frac{p}{q} \cdot m = \frac{a}{b}$$

Solve for m :

$$m = \frac{a \cdot q}{b \cdot p}$$

Since integers are closed under multiplication, $a \cdot q$ and $b \cdot p$ are integers.

However, this implies that m can be expressed as a ratio of two integers, which contradicts the assumption that m is irrational.

Therefore, the product of a nonzero rational number and an irrational number must be irrational. ■

Question 6

Prove that if x is rational and $x \neq 0$, then $\frac{1}{x}$ is rational.

If x is rational then there exist integers p and q such that $x = \frac{p}{q}$, where $q \neq 0$.

Given that q is nonzero, if $p = 0$, $x = \frac{0}{q} = 0$. Thus, by contrapositive, if $x \neq 0$, then $p \neq 0$. Since we have assumed that $x \neq 0$, we can conclude that $p \neq 0$.

Given that $x = \frac{p}{q}$, then $\frac{1}{x} = \frac{1}{\frac{p}{q}} = \frac{q}{p}$.

Since p and q are integers, and $p \neq 0$, $\frac{q}{p}$ is a rational number.

Therefore, if x is rational and $x \neq 0$, then $\frac{1}{x}$ is rational. ■

Question 7

Prove that if m and n are integers and mn is even, then m is even or n is even.

For the sake of a contrapositive proof, assume that the negation of the conclusion implies the negation of the hypothesis:

Since the original conclusion is a disjunction, its negation is a conjunction of the negations of the individual statements. In other words, the negation of “ m is even or n is even” is “ m is odd and n is odd”.

If m and n are integers and both are odd, then they can be expressed as $m = 2a + 1$ and $n = 2b + 1$ for some integers a and b .

Then, the product of m and n is:

$$mn = (2a + 1)(2b + 1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$$

Since integers are closed under multiplication and addition, then $2ab + a + b$ is an integer and $2(2ab + a + b)$ is therefore even. Thus, $2(2ab + a + b) + 1$ is odd.

Therefore, if m and n are integers and both are odd, then mn is odd.

By contrapositive, if mn is even, then either m is even or n is even.

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Question 8

Prove that if n is an integer and $3n + 2$ is even, then n is even using

a) a proof by contraposition The contrapositive of the statement “if $3n + 2$ is even, then n is even” is “if n is not even, then $3n + 2$ is not even”.

If n is not even, then n is odd. Therefore, there exists an integer m such that $n = 2m + 1$.

Substitute $n = 2m + 1$ into the equation $3n + 2$:

$$3(2m + 1) + 2 = 6m + 3 + 2 = 6m + 4 + 1 = 2(3m + 2) + 1.$$

Since $3m + 2$ is an integer, $2(3m + 2)$ is even. Thus, $2(3m + 2) + 1$ is odd.

Therefore, if n is not even, then $3n + 2$ is not even. ■

b) a proof by contradiction Assume for the sake of contradiction that n is an integer and $3n + 2$ is even, but n is odd.

Since n is odd, there exists an integer m such that $n = 2m + 1$.

Substitute $n = 2m + 1$ into the equation $3n + 2$:

$$3(2m + 1) + 2 = 6m + 3 + 2 = 6m + 5 = 2(3m + 2) + 1.$$

Since $3m + 2$ is an integer, $2(3m + 2)$ is even. Thus, $2(3m + 2) + 1$ is odd.

This contradicts the assumption that $3n + 2$ is even.

Therefore, if n is an integer and $3n + 2$ is even, then n is even. ■

Question 9

Prove the proposition $P(1)$, where $P(n)$ is the proposition “If n is a positive integer, then $n^2 \geq n$.” What kind of proof did you use?

To prove the proposition $P(1)$, we need to show that if $n = 1$ is a positive integer, then $1^2 \geq 1$.

Since $1^2 = 1$, and $1 \geq 1$, the proposition $P(1)$ is true. ■

The proof used here is a direct proof, as it directly shows that the proposition holds for the specific case of $n = 1$.

Question 10

Show that if you pick three socks from a drawer containing just blue socks and black socks, you must get either a pair of blue socks or a pair of black socks.

For the sake of contradiction, assume that it is possible to pick three socks from the drawer and not get a pair of blue socks or a pair of black socks.

If there is no pair in this 3-sock selection, then the three socks must all be different colors. In order for the three socks to all be different colors, there must be at least 3 different colors of socks in the drawer. Since there are only two colors of socks in the drawer (blue and black), this is not possible.

Therefore, it is not possible to pick three socks from the drawer and not get a pair of blue socks or a pair of black socks. In other words, if you pick three socks, you must get either a pair of blue socks or a pair of black socks. ■

Question 11

Prove that if n is a positive integer, then n is even if and only if $7n + 4$ is even.

To prove that if n is a positive integer, then n is even if and only if $7n + 4$ is even, we first need to prove the forward implication: “if n is even, then $7n + 4$ is even”.

Let n be an even positive integer. Then, there exists an integer k such that $n = 2k$.

Substitute $n = 2k$ into the equation $7n + 4$:

$$7(2k) + 4 = 14k + 4 = 2(7k + 2)$$

Since integers are closed under multiplication and addition, $7k + 2$ is an integer, and $2(7k + 2)$ is therefore even. Therefore, if n is even, then $7n + 4$ is even.

Next, we need to prove the reverse implication: “if $7n + 4$ is even, then n is even”.

To form a proof by contrapositive, we assume the negation of the conclusion: “if n is odd, then $7n + 4$ is odd”.

If n is odd, then there exists an integer m such that $n = 2m + 1$.

Substitute $n = 2m + 1$ into the equation $7n + 4$:

$$7(2m + 1) + 4 = 14m + 7 + 4 = 14m + 10 + 1 = 2(7m + 5) + 1$$

Since integers are closed under multiplication and addition, $7m + 5$ is an integer and $2(7m + 5)$ is therefore even. Thus, $2(7m + 5) + 1$ is odd.

Therefore, by contrapositive, if $7n + 4$ is even, then n is even.

Since both the forward and reverse implications have been proven, we can conclude that if n is a positive integer, then n is even if and only if $7n + 4$ is even. ■

Question 12

Prove that $m^2 = n^2$ if and only if $m = n$ or $m = -n$.

To prove that $m^2 = n^2$ if and only if $m = n$ or $m = -n$, we need to prove both directions of the statement.

First, we prove the forward implication: "if $m = n$ or $m = -n$, then $m^2 = n^2$ ".

In the first case, if $m = n$, then $m^2 = n^2$.

In the second case, if $m = -n$, then $m^2 = (-n)^2 = (-1 \cdot n)^2 = (-1)^2 \cdot n^2 = 1 \cdot n^2 = n^2$.

Therefore, if $m = n$ or $m = -n$, then $m^2 = n^2$.

Next, we prove the reverse implication: "if $m^2 = n^2$, then $m = n$ or $m = -n$ ".

Given that $m^2 = n^2$, we can rewrite this as $m^2 - n^2 = 0$.

Using the difference of squares formula, we can factor this as $(m - n)(m + n) = 0$.

This equation holds true if either $m - n = 0$ (implying that $m = n$) or $m + n = 0$ (implying that $m = -n$).

Therefore, if $m^2 = n^2$, then $m = n$ or $m = -n$.

Since both directions of the statement have been proven, we can conclude that $m^2 = n^2$ if and only if $m = n$ or $m = -n$. ■

Question 13

Show that these three statements are equivalent, where a and b are real numbers: (i) a is less than b , (ii) the average of a and b is greater than a , and (iii) the average of a and b is less than b .

The average of two real numbers a and b is $\frac{a+b}{2}$.

(ii) implies (i): If the average of a and b is greater than a , then $\frac{a+b}{2} > a$.

Multiplying both sides by 2, we get $a + b > 2a$.

Subtracting a from both sides, we get $b > a$.

Therefore, if the average of a and b is greater than a , then a is less than b .

(i) implies (iii): If a is less than b , then the average of a and b is $\frac{a+b}{2}$.

Add b to both sides of the inequality $a < b$ to get $a + b < 2b$.

Divide both sides by 2 to get $\frac{a+b}{2} < b$.

Therefore, if a is less than b , then the average of a and b is less than b .

(iii) implies (ii): If the average of a and b is less than b , then $\frac{a+b}{2} < b$.

Multiplying both sides by 2, we get $a + b < 2b$.

Subtracting b from both sides, we get $a < b$.

Therefore, if the average of a and b is less than b , then the average of a and b is greater than a .

Since we have shown that (ii) implies (i), (i) implies (iii), and (iii) implies (ii), and implication is transitive, we can conclude that the three statements are equivalent. ■

Question 14

Show that these statements about the real number x are equivalent: (i) x is rational, (ii) $x/2$ is rational, (iii) $3x - 1$ is rational.

(i) implies (ii): If x is rational, then there exist integers p and q such that $x = \frac{p}{q}$, where $q \neq 0$.

$$\text{Then, } \frac{x}{2} = \frac{p}{2q}.$$

Since integers are closed under multiplication, $2q$ is an integer. Therefore, $\frac{p}{2q}$ is a ratio of two integers, making $\frac{x}{2}$ rational. Therefore, if x is rational, then $\frac{x}{2}$ is rational.

(ii) implies (iii): If $\frac{x}{2}$ is rational, then there exist integers a and b such that $\frac{x}{2} = \frac{a}{b}$, where $b \neq 0$.

$$\text{Then, } x = 2\left(\frac{a}{b}\right) = \frac{2a}{b}.$$

Substitute $x = \frac{2a}{b}$ into the equation $3x - 1$:

$$3\left(\frac{2a}{b}\right) - 1 = \frac{6a}{b} - 1 = \frac{6a-b}{b}$$

Since integers are closed under addition, $6a + (-b)$ is an integer. Therefore, $\frac{6a-b}{b}$ is a ratio of two integers, making $3x - 1$ rational. Therefore, if $\frac{x}{2}$ is rational, then $3x - 1$ is rational.

(iii) implies (i): If $3x - 1$ is rational, then there exist integers m and n such that $3x - 1 = \frac{m}{n}$, where $n \neq 0$.

$$\text{Then, } 3x = \frac{m}{n} + 1 = \frac{m+n}{n}.$$

$$\text{Divide both sides by 3 to get } x = \frac{m+n}{3n}.$$

Since integers are closed under addition and multiplication, $m+n$ and $3n$ are integers. Therefore, $\frac{m+n}{3n}$ is a ratio of two integers, making x rational. Therefore, if $3x - 1$ is rational, then x is rational.

Since we have shown that (i) implies (ii), (ii) implies (iii), and (iii) implies (i), and implication is transitive, we can conclude that the three statements are equivalent. ■

Question 15

Show that the propositions p_1 , p_2 , p_3 , and p_4 can be shown to be equivalent by showing that $p_1 \leftrightarrow p_4$, $p_2 \leftrightarrow p_3$, and $p_1 \leftrightarrow p_3$.

Since $p_2 \leftrightarrow p_3$ and $p_1 \leftrightarrow p_3$ and bidirectional implication is transitive, we can conclude that $p_1 \leftrightarrow p_3 \leftrightarrow p_2$ which implies that $p_1 \leftrightarrow p_2$.

Since $p_1 \leftrightarrow p_4$ and $p_1 \leftrightarrow p_3$ and bidirectional implication is transitive, we can conclude that $p_3 \leftrightarrow p_1 \leftrightarrow p_4$ which implies that $p_3 \leftrightarrow p_4$.

Thus, when $p_1 \leftrightarrow p_4$, $p_2 \leftrightarrow p_3$, and $p_1 \leftrightarrow p_3$, it's also implied that $p_1 \leftrightarrow p_2$ and $p_3 \leftrightarrow p_4$, and by transitivity, we can conclude that $p_1 \leftrightarrow p_2 \leftrightarrow p_3 \leftrightarrow p_4$ and the propositions p_1 , p_2 , p_3 , and p_4 are therefore equivalent. ■